

Software Requirements Specification (SRS)

Project: Banking Website

(Note: Highlighted Text = Revisions Made)

1. Introduction

1.1 Purpose

The purpose of this document is to outline the functional and nonfunctional requirements for a banking website. This website will be made for everyone from people to businesses, and will allow users to create wallets/accounts, transfer money, as well as track their transaction history.

1.2 Scope

The banking website will offer a web-based platform for users to manage their finances. The website will provide features such as secure user login, money transfers, and the viewing of transaction history. This website aims to provide a secure and simple banking experience to its users.

1.3 Definitions, Acronyms, and Abbreviations

- **User Roles:** Customers, Employees, Administrators
 - **Money Transfers:** The movement of money between accounts
 - **Transaction:** To buy or sell something
 - **SRS (Software Requirement Specification):** Document that outlines the requirements of a software product
-

2. Overall Description

2.1 Product Perspective

The system will perform as a tool for customers to access banking functionalities without the need to access a local branch. It will allow customers to create incoming and outgoing transfers. Customers will have easy access to balance and transaction history while retaining tight security for full confidentiality of banking information.

2.2 Product Features

1. Account registration, login, and management
2. Creation of multiple wallets per user
3. Display of account balance
4. Incoming transfers will add to wallet balance
5. Outgoing transfers will subtract from wallet balance
6. Display analysis of user spending trends (For example, see Figure 2.2-1)
7. Downloadable account transaction history

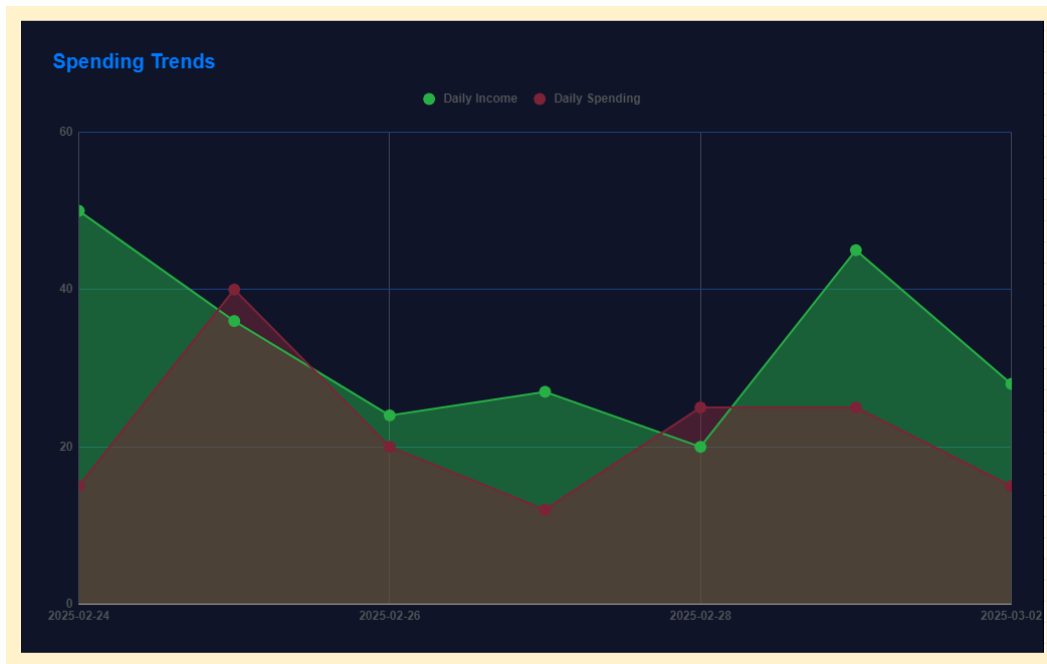


Figure 2.2-1: Spending Trends Graph Example

2.3 User Characteristics

Customers require:

- Basic computer usage abilities

- Computer access

Administrators must:

- Use the system to oversee and manage transactions
- Have the ability to recognize and flag suspicious activity

Employees must:

- Use the system to access an overview of transactions

2.4 Constraints

1. The system will be operated on a web-based interface.
2. The system must be able to handle up to 200 consecutive users

2.5 Assumptions and Dependencies

- All users have access to the internet
-

3. Functional Requirements

- **FR1:** The system shall allow users (admin, employee, and customers) to create an account, obtain member and account numbers, log in and access their respective dashboards.
- **FR2:** The system shall allow users to add funds into their wallets.
- **FR3:** The system shall allow users to use funds from their wallets.
- **FR4:** The system shall track and display users' spending trends under Spend Analysis. (For example, see Figure 2.2-1)
- **FR5:** The system shall allow users to manage multiple wallets.

- **FR6:** The system shall allow users to update their account information under the Account section.
-

4. Non-Functional Requirements

- **NFR1:** The system shall keep a record of transactions so users can track their incoming transfers, outgoing transfers, and spending.
- **NFR2:** The website shall load pages quickly, within a few seconds.
- **NFR3:** The website shall have a simple and easy-to-use interface so that users can navigate without confusion.
- **NFR4:** The website shall require strong passwords for user accounts, where a strong password is one that is resistant to unauthorized access by being difficult to guess or hack.
 - Shall be at least 8 characters long.
 - Shall include at least one numeric digit (0-9).
 - Shall include at least one lowercase letter (a-z).
 - Shall include at least one uppercase letter (A-Z).
 - Shall include at least one special character (!, @, #, \$, etc.).
- **NFR5:** The system shall ensure that users cannot transfer more funds than their available balance.

- **NFR6:** The system shall allow users to download their transaction history in a text document (TXT) format.
-

5. External Interface Requirements

5.1 User Interfaces

- The bank system shall provide a simple, fast, and reliable web application interface.
- The bank system shall support separate dashboards for bank clients, employees, and administrators each with different permissions.
- System shall provide a multitude of services in their own sections including:
 - Card services that displays credit visibility
 - Transaction history
 - Spending analytics in various charts for users
 - User account details
 - Locations of physical locations with contact details
 - Transfer abilities between users and between accounts of a user
 - Employee moderation abilities of above services
 - Administrator moderation abilities of above services

5.2 Hardware Interfaces

- Bank system shall support every day standard desktop and mobile hardware configurations.

5.3 Software Interfaces

- Bank system shall entrust data to a database management system (e.g., MySQL or PostgreSQL).
- System shall accommodate authentication via 2-factor authentication services. (e.g Auth0, Google Firebase Authentication, Duo Security)

- Shall interface with visual analytical services to provide users charts (e.g. Chart.js, D3.js, Power BI, Tableau API)

5.4 Communication Interfaces

- The system shall use HTTPS for secure communication
 - System shall use hashing of sensitive data to prevent data leaks
 - System shall validate user authenticity via 2-factor authentication methods.
-

6. Appendices

6.1 Glossary

- **Balance:** A current total of money held within a bank account
- **Wallet:** A digital storage of different payment methods
- **HTTPS:** Hypertext Transfer Protocol Secure
- **Hashing/Hashmap:** Use of a system to secure data by assigning a key to entered information that is needed for its retrieval